

TOI-2046b

R-2271

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI2046_230726 · created 2026-08-16T22:57:03+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460151.832 +/- 0.048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.027
Transit depth (Rp/Rs)^2	1.98 +/- 0.76 [%]
Orbital Inclination (inc)	88.46 +/- 2.79
Airmass coefficient 1 (a1)	0.917 +/- 0.012
Airmass coefficient 2 (a2)	0.058 +/- 0.032
Scatter in the residuals of the lightcurve fit is	1.31 %
Best Comparison Star	#3 - [64, 393]
Optimal Aperture	3.32
Optimal Annulus	11.53
Transit Duration (day)	0.081 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.141 ± 0.027 vs 0.1213 (deviation 0.68σ)
Duration fit vs published	0.081 d vs 0.1033 d (deviation 0.7σ)
Mid-time O–C	-13.7 min
Depth SNR	4.9
Residual scatter	1.31 %

Light curve

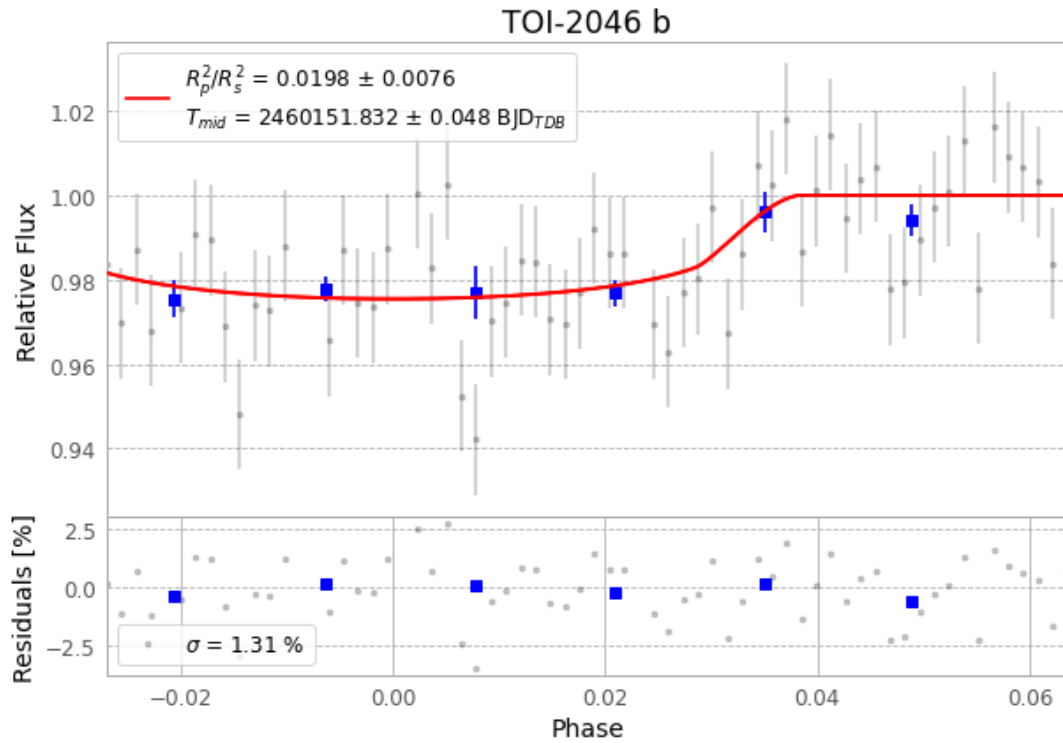


Figure 1 — FinalLightCurve_TOI-2046 b_2023-07-26.png (SHA-256 00037f511bfa588b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [287, 252], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 edf803109ba41a5f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

63 FITS frames (41 MB), TOI2046230726070015.FITS → TOI2046230726101515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI2046-230726.log (001a9f1a6065...), FinalLightCurve_TOI-2046 b_2023-07-26.png (00037f511bfa...), FinalLightCurve_TOI-2046 b_2023-07-26.csv (77c695754f6b...)

Compute resources

Wall time 87.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 22:57Z.